

Yusuf Hasan

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EDUCATION

University of British Columbia

Bachelor of Applied Science, Electrical Engineering

Vancouver, BC

Expected: November 2026

EXPERIENCE

Software Development Engineer Intern

June 2026 – Present

Amazon

Vancouver, BC

- Sole engineer on a **full-stack self-service configuration platform**, adopted by **40+ business clients and internal teams**, giving them visibility into profiles — collections of keys and plugin/selector functions invoked on actions (e.g. subscription creation) — stored across hierarchical datatables with fallback and inheritance
- Designed API contract in **Smithy IDL**; implemented read service in **Java on AWS Lambda** with a **dynamic, cascading lookup** across **7 optional filter dimensions**, achieving **~64 ms** end-to-end reads against $p50 \leq 500 \text{ ms}$ / $p95 \leq 1.5 \text{ s}$ SLOs
- Built client-facing **React + TypeScript** SPA with dependency-aware **cascading dropdowns** and a **pluggable API-client layer**, decoupling frontend delivery from backend readiness
- Provisioned cloud stack as **infrastructure-as-code (AWS CDK)** — Lambda, VPC, Route 53, IAM; hardened with **99+ unit/integration tests**, including schema **drift-guard tests**, shipped via peer-reviewed CI/CD

Hardware Pre-Development Intern

June 2024 – April 2025

Robert Bosch GmbH (Bosch)

Reutlingen, Germany

- Designed **STM32**-based sensor test and data-collection **PCBs** in **Altium** with onboard memory, multi-protocol interfaces (FDCAN, SPI, I²C, UART, USB3300), and an integrated **12V→5V buck converter** with **3.3V LDO** regulation, enabling evaluation of **next-generation MEMS inertial sensors** at $>12.8 \text{ kHz}$
- Wrote efficient **C firmware** to validate sensor functionality and performed data collection and analysis on **10+ automotive MEMS sensors** concurrently during field testing
- Analyzed IMU data in **MATLAB and Python** to characterize next-generation sensor performance (bias, noise, offset, sensitivity) under different vibration/stimulus profiles
- Developed a robust measurement methodology to characterize MEMS gyroscope **offset g-sensitivity** using a modified **rate-table** maytagging approach that accounts for Earth-rotation, implemented in Python

PROJECTS

Hybrid Localization Unit | C, Python, ESKF, PCB Design, IMU, UWB

Sep 2025 – Present

- Designed a **sensor fusion system** combining **UWB ranging** and **inertial measurements** for indoor robotics localization
- Modelled and tuned an **Error-State Kalman Filter (ESKF)** for real-time fusion of IMU and UWB data, implemented in C on a physical hardware module
- **Designed MCU PCB** with a DWM3000 UWB module and MEMS IMUs, and integrated the sensor fusion algorithm

2-DoF Laser Projector System | Altium, Embedded C, SolidWorks, FPGA, Verilog

Feb 2024 – April 2024

- Collaborated in a **team of 4** to design a motor-controlled laser projector system, programming a digital logic decoder in **SystemVerilog** for closed-loop system modelling
- Programmed an **ESP32 interrupt service routine** in C for **SPI** communication and dual motor control, implementing a real-time **PID controller** to project multiple animations

RC4 Decryption and Cryptanalysis Accelerator | SystemVerilog, FPGA, FSM Design

June 2025

- Designed a hardware RC4 stream cipher accelerator in **SystemVerilog**, implementing the **KSA** and **PRGA** across coordinated finite-state machines
- Built an automated **brute-force cryptanalysis system** testing **16.7 million (24-bit) keys** with real-time plaintext validation, optimized for FPGA synthesis with **dual-port memory blocks** and **pipelined state machines**

TECHNICAL SKILLS

Software: C, Java, Python, TypeScript, MATLAB, SystemVerilog, 8051 Assembly, ARM Assembly, React

Cloud & Tools: AWS (Lambda, CDK, VPC, Route 53, IAM), Azure, Git, CI/CD, STM32CubeIDE, Quartus, Simulink, TensorFlow, PyTorch

Electrical: Altium, Cadence Virtuoso, KiCad, LTspice, PSIM, SolidWorks, Logic Analyzer, Function Generator, Oscilloscope